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Model for Faculty, Student, and Practitioner Development in Sustainability Engineering through an Integrated Design Experience

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Abstract: Sustainable development and the green building movement have been adopted faster than any recent movement in the engineering field. With over 40% of the total U.S. energy usage servicing the operation of commercial and residential buildings, this trend is well founded. Recent surveys of the industry indicate that within 4 to 5 years, a vast majority of engineering firms expect their business will be significantly dedicated to green building designs. In contrast, current academic institutions are not well positioned to prepare young engineers for this challenge, and current faculty are not well trained in the tenets of sustainability or the roles of engineers in this movement. Change must occur if the engineering and design professions are to remain relevant and responsive to societal needs. To accommodate this challenge, the writers have designed and implemented the Integrated Design Experience (IDeX), a capstone course in which undergraduate and graduate students interact with faculty and practitioners on real projects with challenging needs in sustainability. The course is designed to provide an actual and virtual space for the multitude of disciplines to interact on real designs to foster both improved research and outreach efforts. Expected outcomes from the course include both student and faculty learning on the methods and value of sustainable design as well as the development of an interdisciplinary network of faculty and practitioners involved in sustainable design. Learning is being evaluated using a continuous authentic assessment of design products. First-year results indicate that students learned interdisciplinary teamwork and communication skills, and they see substantial value in the authentic design experience. In future years, the development of the interdisciplinary network will be tracked by using social networking tools and by assessing faculty attitudes toward involvement in IDeX. Both metrics will be investigated using the diffusions of innovation framework. The combined evaluation will lead to an in-depth understanding of how the IDeX model can be scaled and replicated at other institutions. **DOI: 10.1061/(ASCE)EI.1943-5541.0000058.** *© 2011 American Society of Civil Engineers.*

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Introduction

Educating students about sustainability and sustainable engineering design requires careful consideration of the situated knowledge essential to undergraduate and graduate students becoming proficient in the field. Educators must be able to teach students the ways to access resources to develop their expertise. Experts in learning suggest that most or all knowledge is situated in a specific context; therefore learning outside of an applied context has minimal effectiveness (National Research Council 1999). The dynamic contemporary study of sustainable engineering design requires close attention to how multidisciplinary practitioners of sustainable design approach design problems in their respective fields. The goal of this paper is to propose a model that directly addresses the need for situated learning along with the need for faculty and practitioners to be motivated to integrate with different disciplines and participate in a sustainable design experience.

The need for sustainability education is directly linked to changes in contemporary practice that respond to fundamental societal demands for more responsible design. The need is not founded in any one discipline but is critical to the challenges faced in every discipline; the need for sustainability design relates to current demands for energy efficient buildings, improved storm water handling, efficient transportation networks, and so on. This claim is supported by recent studies that found the adoption rate of green building practices so rapid during the last decade that 94% of current architectural and engineering firms "expect to be significantly dedicated to green (designs) by 2013" (Bernstein and Bowerbank 2008). Yet these same firms consistently report that engineering graduates lack even the most fundamental understanding of sustainability and sustainable engineering design to respond to current needs.

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Anecdotal evidence suggests that there is substantial student interest, especially among female students, as judged by proportionally high female enrollments in sustainable design courses at Washington State University (WSU). Although some colleges and universities have implemented engineering courses on sustainability, there is no widespread infusion of the underlying concepts into engineering curricula. Efforts are hindered by (1) insufficient knowledge of sustainable design practice by academics, (2) insufficient research basis for many sustainable designs used by practitioners and, (3) lack of a model depicting how our current programs can transition sustainable design concepts into the curriculum.

Efforts currently exist to incorporate sustainability into engineering education. One of the most notable is the Center for Sustainable Engineering (CSE) (Davidson et al. 2010), a collaboration of faculty from several universities supported by the National Science Foundation (NSF) and EPA. CSE's current goal is to develop and implement activities to enhance education in sustainability engineering at colleges and universities worldwide. Some specific activities include workshops to aid faculty to join a network of educators in sustainability engineering website development to provide educational materials and an assessment of current sustainability engineering courses as a benchmark. CSE's activities should provide many of the tools to develop a sustainability engineering curriculum but do little to overcome a major challenge: a lack of sustainability engineering knowledge among current academics. The Integrated Design Experience (IDeX) described in this paper attempts to augment and validate the CSE work while addressing the root cause impeding adoption.

The objective of this paper is to (1) demonstrate how an integrated design experience can address practical and theoretical barriers to incorporating sustainability concepts into the curriculum, (2) describe how the effectiveness of IDeX will be determined, and (3) share results on the impact of IDeX on student learning. IDeX incorporates faculty from engineering, architecture and the creative design disciplines, construction management, and practitioners in a variety of relevant professions into a year-long capstone design course focused on the development of authentically situated sustainable design knowledge in transdisciplinary undergraduate and graduate students.

What Is Sustainable Design and Why Is Integrated Design Appropriate?

In sustainable design, the goal is to redefine the relationship of a project with its environment and users (McLennan 2006). The approach strives to increase efficiencies in operation and construction of infrastructure, thereby decreasing resource dependence, while improving resource-conscious behavior of users. Many tools are available to facilitate such designs, including green building standards like LEED or design methodologies like the Natural Step Framework. In addition, design goals such as carbon neutral construction, NetZero Homes, and Passivhaus are emerging to push the techniques of energy efficiency. Resource independence is taken a step further in the Living Building Challenge that strives for both energy and water independence along with healthier materials, whereas the concepts of building constructability and reuse are stretched in the Life Cycle Building Challenge. But regardless of the method or the goal, real change is always facilitated by a holistic approach known as *integrated design*.

Integrated design is an approach that engages a multitude of disciplines to collectively drive the design to new levels (Thibaudeau 2008). The technique can facilitate the design by using established green building standards as a guide (e.g., LEED, National Green Building Standard, Green Globes, Sustainable Sites) but is not necessarily driven by such processes. Rather, integrated design only uses a green building standard as a tool to achieve a sustainable outcome if it is deemed necessary by local, state, or federal regulation; is useful as an organizational process for the team; or is important to the owner for marketing the facility. In general, green building standards can be useful because they approach building practices in a holistic manner seeking to derive changes from multiple aspect of design (Yudelson 2009).

In the integrated design process, designers and the developer or owner conceive the overall project concepts together. Designers configure the site and landscape so that the development fits as seamlessly as possible into natural features of the site, minimizes its impact, and manages storm water through natural flow patterns for maximal soil retention. The design team positions the building on the site to minimize excavation disturbance and maximize tree retention. The positioning and design of the building strive for aesthetics while optimizing solar gains for day lighting and heating/cooling requirements. Designers employ efficient mechanical systems that minimize energy and water usage to meet the reduced requirements provided by passive solar gains and water retention. In addition, they make material choices that maximize the life of the project and minimize both material waste during the construction process and energy waste in the production of the materials. In selecting materials, designers consider the inherent toxicity through off-gassing and potential consequences of materials in catastrophic situations like fire. Building system choices are made that minimize material waste during construction and maximize the life of the structure. Finally, the design must meet life-safety codes that dictate structural and fire safety requirements while minimizing material usage and sourcing materials to meet the broader environmental goals. Using integrated design, each of these steps is not performed autonomously but in concert to accommodate competing needs. Thus, the design process proceeds in a far more efficient manner than other models for design practice.

Barriers to Integrating Sustainability into the Curriculum

The integration of sustainability into an engineering curriculum is challenging because of a multitude of factors, including limited knowledge by engineering faculty about sustainability and sustainability engineering, a lack of resources to develop baseline proficiency, a lack of models of successful integration, and a lack of incentives for the required effort. Using a holistic and integrated view of sustainability is commonplace in practice but rare in engineering education, yet it is essential to prepare students for developing innovative solutions to design challenges in practice.

Engineering faculty commonly do not have experience as practicing engineers and have limited or no knowledge of current design practices, especially in rapidly changing fields like sustainable engineering design. Clearly, this hinders the ability of faculty to teach about sustainable engineering design at the same time limiting the availability of sustainability courses for students. Compounding the lack of expertise on engineering design by engineering faculty is a lack of relevant and accessible resources for developing expertise. Although integration of practitioners into capstone design courses is commonplace, roles are often limited and practitioner incentive for participation is not directly addressed. Practitioner involvement is motivated by stated needs by numerous local firms for students trained in sustainable design in their higher education experience.

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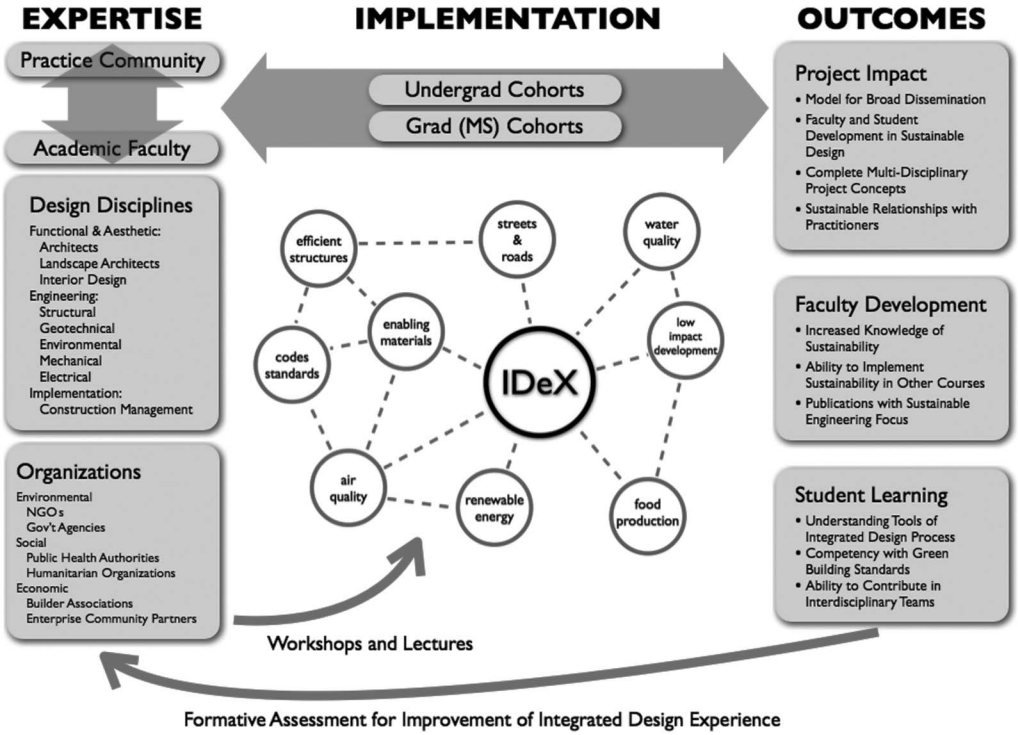

Fig. 1. Schematic diagram of the IDeX currently being deployed

Even if resources were available for faculty to develop expertise in sustainable engineering design, they must be motivated to do so. Faculty members have active and demanding research agendas that are at the core of research universities' promotion and tenure requirements. The IDeX program (Fig. 1) attempts to address this concern by providing a structure for faculty to mentor Masters students who are funded through fellowships and publish in peer-reviewed journals during the mentoring experience.

Cognitive Apprenticeship Approach to Faculty and Student Development

Considering theoretical models of cognitive apprenticeship provide a structure to develop faculty competence in sustainability through an integrated learning experience involving faculty, experienced engineering and architecture practitioners, and undergraduate engineering students. Cognitive apprenticeship is a learning theory that describes the process by which an apprentice learns the practice, behavior, and actions of a master in the field through development of a shared understanding of that field (Collins et al. 1987). This model is particularly relevant and appropriate for the proposed path to faculty and student development because faculty often lack the knowledge of sustainable design, which is commonly practiced by knowledgeable practitioners. The three most relevant and important components of cognitive apprenticeship to this model are the tacit processes implicit in design but often missing from traditional instruction, the value of coaching, and the authenticity of the design problems (Edmondson 2005).

Tacit knowledge and processes are those that are embedded in day-to-day practice but difficult to portray through formal instruction. These processes are best addressed through continuous interactions between novice and expert. Faculty and students will become familiar with the processes of sustainable design in an efficient and effective manner through continuous interaction with, and feedback from, sustainability practitioners, where faculty and students can practice and improve these processes. As quoted by Edmondson, cognitive apprenticeships "are designed, among other things, to bring these tacit processes into the open, where students can observe, enact, and practice them with help from the teacher." (Collins et al. 1987).

Coaching of faculty and students occurs throughout the project design process. Since coaching is accomplished on the basis of formal and informal social interactions, it will focus directly on existing knowledge of faculty and students. Coaching in the integrated design experience will include continuous feedback, and improved modeling of expert behavior, with the long-term goal of faculty and students benefiting from their interactions with the expert sustainable designers.

Finally, and perhaps most importantly, the skills of sustainable design will be authentic and in context by implementing real-world, community sponsored, and expert mentored design projects with challenging sustainable design goals. Collins et al. (1987) argue that cognitive apprenticeships are less effective when skills and concepts are taught independent of their real-world context and situation. As they state, "Situations might be said to coproduce knowledge through activity. Learning and cognition, it is now possible to argue, are fundamentally situated." (Brown et al. 1989). In cognitive apprenticeships, the activity being taught is modeled in real-world situations (Edmondson 2005).

Integrated Design Experience Approach to Sustainability Education

The long-term goal is to develop a model for integrating concepts of sustainability into the curriculum that can be applied at a diverse group of institutions and settings. The writers have created a network-style (IDeX) for undergraduate and Masters students using a team composed of engineering, architecture and the creative design disciplines, construction managers, and practitioners from a variety of relevant professions who use or impact the built environment. This model is unique because it seeks to develop a

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dynamic interaction between university faculty and professionals who exchange expertise to enhance the transdisciplinary design experience.

Simultaneously, the writers are working to increase existing faculty expertise in sustainability through multiple pathways including:

- Curriculum development in conjunction with faculty and professional sustainability experts;
- Integrated design experiences that lead to research projects and collaborative groups of faculty;
- Workshops for knowledge on sustainability and on sustainable design.

Curriculum Design and Implementation Plan

The cornerstone of this sustainable design effort is to develop real projects that aggressively push the concepts of resource efficiency while facilitating sustainable lifestyles. To implement the integrated design program, cohorts of undergraduate (BS) and graduate (MS) students from a variety of disciplines have been established. The initial project has been a sustainable design of the Washington State University organic farm: a project called smartFARM. The student design team consisted of 24 students from architecture, landscape architecture, construction management, and organic agriculture, as well as mechanical, electrical, environmental, and civil engineering. In different projects, these groups might be augmented by similar level students from a variety of other disciplines depending on the project requirements. Examples include health care (nursing, physiology, or pharmacy), computer science (human interface development), business, public policy, and so on.

A number of overarching tenets and techniques have been used to develop the course and are employed throughout.

Open-Source

A basic hypothesis of this project is that the design methods used for infrastructure and codes regulating this design have evolved in an open-source manner using a transparent consensus process that allows for broad input of ideas to balance needs. Unlike proprietary design used in the myriad of products that society employs, the accepted methodology employed to design infrastructure requires following methods developed by approval and input of the larger professional community. This accepted, open-source process is most explicitly realized in building codes but is evident in codes governing the design of other infrastructure; from storm water to fire to health standards. Therefore, influencing this consensus process requires adopting an attitude that all of the design work is open and transparent to anyone that is interested. This philosophy can be seen in the course weblog, IDeXstudio.org, the solicitation of the professional design community, and more subtly in the organizational structure of the project. The open-source movement has been most recently and overtly adopted by computer software developers who have used it to develop products such as the Linux operation system, Firefox Web browser, and the Wikipedia encyclopedia to name just a few. In the last decade, the philosophy has made its way into a variety of other design fields (Williams and Anderson 2007; Panchal and Fathianathan 2008) but is just now being discussed in the field of architecture ("Community-focused" 2010; Hight and Perry 2006).

Design Professionals

The Pacific Northwest is a leader in sustainability; Portland and Seattle are assessed as numbers 1 and 3, respectively, on the Sustain Lane 2008 U.S. city rankings ("2008 US" 2008). The professional design firms in the region have a strong impact worldwide. Because of their importance, numerous prominent firms in other regions have also opened branch offices in the Pacific Northwest. Together these firms support experienced government agencies and are allied with nongovernmental organizations that offer a wealth of programming dedicated to realizing a sustainable society. To assist development and adoption, this professional community has a strong role in the overall process. First and foremost, their role is to assist by providing their experience and knowledge into the open-source design process. This component is crucial because students often lack the experience of best practices and implementation that exists in professionals. Likewise, the design community provides a broader network of experts on which to draw for the overall progression of the project. The expert knowledge, creativity and latitude provided by the academic community provide a synergistic component pushing creative solutions.

Interaction with sustainability experts occurs in a variety of ways; regular on-campus guest lectures and seminars are the most common. Examples of professional development through sustainability experts are programs offered by the Northwest Earth Institute (NWEI). The NWEI is a nationally recognized nonprofit organization that develops programs to enable organizations to develop sustainability solutions tailored to their specific professions. Because the built environment is the very environment that facilitates the writers' behavior, these is particular interest in NWEI's choices for sustainable living (CSL) module. Organized as a series of readings and self-facilitated discussions and exercises, CSL is aimed at developing a broader sense of sustainability goals. The exercises are meant to develop sensitivity toward the role that the design will play in facilitating or hampering sustainable habits in the owners. The module will encompass approximately seven 1 h sessions and will be implemented during the initial part of the first semester.

The integrated design process involves a collaborative and iterative method of engaging all of the stakeholders simultaneously in the project design. Though quite flexible, the method has specific goals and techniques to foster a successful outcome. Since the process is collaborative, both students and faculty must understand the details to realize a successful mentor relationship. Many firms in the Pacific Northwest region are well founded in the integrated design process and are sought to assist participants in understanding a successful process. Students and faculty visit the firms for specialized seminars and presentations on challenges to implementing sustainable principles in design, new technologies, and successful projects.

Finally, social networking strategies like Facebook, Twitter, and Flickr are employed to raise the visibility of the efforts and reach past immediate personal professional networks. The effort, although modest and in its infancy, has developed hundreds of unique viewers and followers. In the long term, the effort may provide valuable input by building a network of professionals and experts outside the immediate region and WSU alumni.

Integrated Course Format and Structure

The education methodology for this year-long course consists of lectures, discussions, guest speakers, seminars, and a design studio. The students are evaluated using a series of weblog and social network postings, reports, homework, and presentations. The general framework for IDeX is diagramed in Fig. 2. This process is represented as a clear linear flow, however, it is important to understand that the process is never so well defined. Rather, it is expected that each project and class will differ, necessitating that flexibility and adaptation be crucial components of the process. The course is structured as follows: early in the process students are divided into multidisciplinary teams focused on the pertinent resource flows.

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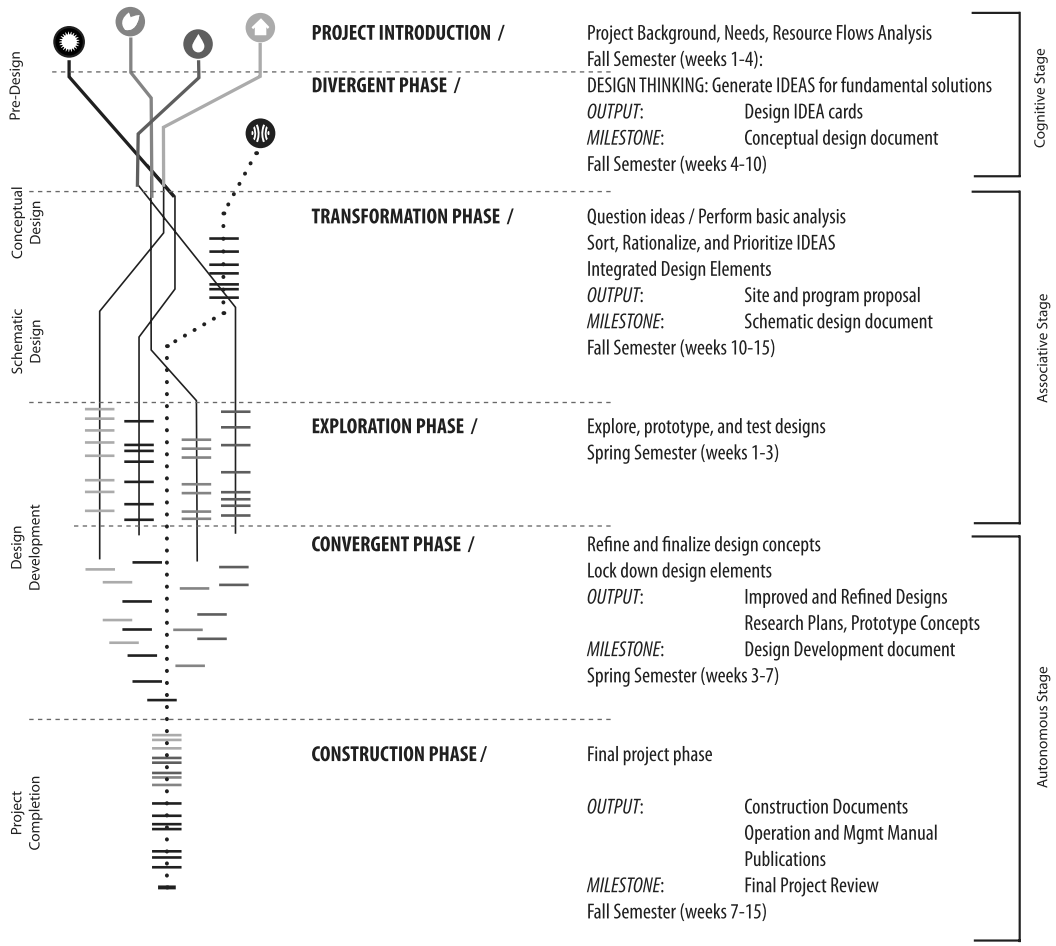

Fig. 2. Diagram of the IDeX timeline depicting various learning and design phases with the expected outcomes and milestones

Faculty members and outside professionals float between all the teams acting as omnibus advisers. The teams participate in a pre-design charrette to determine the particular design strategies to adopt. The teams meet as one large group weekly for lectures, seminars, and group discussions of relevant issues. This combined weekly meeting also serves as the forum for general education in sustainability engineering issues and technology. Subsequent meetings allow each team time to discuss the new issues as they pertain to that team's design problem and to meet with faculty in their individual disciplines. The teams have access to a project-specific site online that allows continuous interactive input: uploading of research material, suggestions for new design directions, design changes in drawings, and so on. Online material is printed regularly and reviewed by the team as well as teams of invited professionals.

The project begins with a month-long project introduction phase where instructors review the project background, basic client needs and vision, and end with a resource flows analysis that provides the team with an intimate vision of the operations and challenges that must be addressed. Participants develop a design manifesto as a reminder of their goals and approaches. During this time, participants also develop basic communication skills among the different groups by introducing language and terminology that will help them speak effectively to one another. At the same time, instructors explain the different strengths, approaches, and methodologies employed by the various disciplines. There is a need to develop a common language with which the groups can communicate; the common language can be developed on the basis of basic diagramming and analytical skills.

In the transformation phase it is necessary to sort, rank, and rationalize the many different design ideas. The prioritized group of design ideas forms the basis for a conceptual design of the project. Ideas can be regrouped, combined, and more thoroughly analyzed to help the conceptual design mature into a schematic design. The participants formalize this milestone in a document that undergoes a preliminary and final review and serves as an important component of evaluation.

The following semester begins with steps that lead toward design development. In the exploration phase, the design team must test, analyze, and refine the initial ideas put forth in the schematic design document. Participants work in resource-specific groups during this time, interacting and seeking counsel from peers to test ideas. After an outside review process, the team can then move into the convergent phase in which design concepts and ideas become fully developed and finalized. This phase ends with a design development document that undergoes a formal review for evaluation.

In the final stages of the project, participants break into disciplinary teams to develop the final project outcomes that include items such as component prototypes; testing, evaluation, and engineering reports; operations and management manuals; construction documents; and design and research publications.

Examining Project Success

The IDeX program is designed to provide a sustainable and authentic means for integrating undergraduate and graduate students, faculty and practitioners (Fig. 1). As a result of the IDeX, the writers

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expect involved student and faculty to develop knowledge of sustainable design in authentic contexts and for students to develop interdisciplinary teamwork and communication skills. It is also expected that participants will develop a functional network of faculty and practitioners that have incentive to participate in the IDeX program over the long term. Assessment of faculty and student learning is being accomplished through a study employing longitudinal mixed methods focusing on situated knowledge in design. Also being examined are the development of a functional network by using the tools of organizational network analysis and the incentive to participate by using the framework of diffusion of innovation. Analysis of the first-year IDeX cohort has focused on student learning and on participating faculties definition of IDeX (i.e., their innovation configuration).

Student Learning

The IDeX experience is a problem-based learning environment, where students interact with a design project that changes annually. Problem-based learning environments do not lend themselves to content-specific outcomes, such as the ability to understand stresses in beams and columns. However, they are useful in delineating broad design and attitudinal outcomes, which might be expected from course structures like IDeX. Students are expected to design a project using sustainable design principles in an interdisciplinary environment, to communicate effectively, and to contribute as team member within an interdisciplinary design environment. Sustainable design principles include competency with green building standards. The student's ability to design the project is assessed by continuously evaluating them through periodic design reviews and formal design critiques. Assessment is conducted by both faculty and peer assessment that focuses on the value and contribution of fellow IDeX students.

Assessing teamwork and communication skills and the real-world nature of the projects was conducted through extensive qualitative student interviews. In the long term, a set of validated instruments originally designed for senior capstone engineering courses will be used to assess these skills (Beyerlein et al. 2006; Davis et al. 2006). These instruments are formative in nature and allow for evaluating continuous improvement of performance by students over the course of the term.

A series of individual interviews was conducted to (1) evaluate students' perceived value of participating in the IDeX course and (2) develop a holistic view of how the students have defined their interactions with the course. Students perceptions of the course are vital for newly developed programs like this because they provide insight into the course value for future cohorts. These perceptions have implications for recruiting and retaining new students in the course and for the transferability of this course to different settings. The interviews were conducted in the last few months of the course with 13 of the 26 students. This sample of students represents the entire cohort from the basis of academic achievement and gender. The questions asked were developed by a group of faculty members who participated in the IDeX course and refined by the group of graduate students who administered the interviews. The interview questions were grouped into general categories: (1) the student's views on the interdisciplinary interactions and authentic qualities of the course and (2) how sustainable design was integrated into the course. The students' names were not recorded for anonymity purposes. Their ages ranged from 21 to 53 years and they studied in the fields of civil engineering, architecture and organic agriculture. Seven students were pursuing undergraduate degrees, whereas the others were in graduate programs.

None of the interview questions directly addressed communication and teamwork, but the students often spoke about these qualities. Every one of the 13 interviewees referred in some way to communication and teamwork in the class. Interviewee 2 shared what he learned from interdisciplinary interaction: "I learned a lot about how to talk with other people. Learning the language and trying to explain not my issues, but how to explain structural engineering concepts to them…I just think communication was the main thing I learned." When responding to the same question, interviewee 5 said "I think I learned that there's a lot of stuff I don't know and instead of me going out and learning that stuff on my own it's more beneficial for a team to have someone who's like an expert in each thing and you rely on those other people instead of relying on your own learning." When interviewee 1 was asked what interdisciplinary interactions would aid in her future disciplinary work she responded with "The main thing was how to communicate with other people, kind of how to get your ideas across and how to listen to them and their input and how to incorporate it into what you know…" Seven of the 13 students responded similarly to that question. The communication and teamwork skills that the students' gained from the course were prevalent in their responses.

The IDeX course was intended to be an authentic experience, and most of the students felt that it was. Twelve of the 13 students interviewed mentioned the authenticity of the experience. When asked if he would recommend the IDeX program to other students, interviewee 3 responded, "Overall I would say yes because it is a real-life experience that will help you learn how to interact with people outside of engineering. It won't be as hard in the real world if you had it in school." The students were also asked if they felt the course prepared them to address real-world situations. Interviewee 1 said, "Yea for the most part. Taking the blank slate and pretty much trying to develop it into the needs of the client and setting restrictions or, in our case, university restrictions as far as development, codes, and developing the site." The response from interviewee 3 was, "Yes because we actually got to work on a real-life project and learning the steps and working toward a final design. Other classes I've taken haven't been this detailed. Interviewee 8 highlighted the benefits of interdisciplinary interactions: "Some of the benefits, I think it's been really eye-opening to like maybe real-world practice and some of the hurdles that we'll have to experience." The students felt that the IDeX course was authentic and helped prepare them for future career experiences. As the new IDeX project began this year, alumni of the course directed many of the new students to it. In addition, approximately one-quarter of the alumni on campus this semester attended the opening class to learn about the new project and volunteer their assistance. Although anecdotal, the enthusiasm for the course that continues through the following school year is encouraging.

Faculty Learning

Our assessment of a faculty's ability to develop knowledge of sustainable design centers on their ability to implement sustainable design in their courses and to produce publishable practice journal articles on sustainable design in their fields. In other words, it is not nearly as important to determine what they know as it is to make certain that they feel they know enough to implement what they learn through interaction with IDeX in their other courses and research activities. Evidence of this implementation and publication productivity is assessed using one-on-one interviews with faculty in conjunction with interviews on diffusion of innovation described subsequently. No publications resulted from year-one activities, but manuscripts are in preparation and may be published in the upcoming year. Some of these publications are resulting from six nonthesis M.S. projects that directly resulted from IDeX. A total of nine faculty participated in advising these students, most

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of who did not previously participate in sustainable design research.

Organizational Network Analysis

It is expected that over the course of implementing the IDeX program, networks of practitioners and faculty will develop through both the purposeful interactions that are part of the formal IDeX program and the informal and serendipitous interactions fostered by the program. For example, none of WSU faculty members working directly with IDeX worked together on a substantial project prior to the start of the program. However, those same faculty currently collaborate outside the course framework on a regular basis. This collaboration has resulted in student research as well as proposals to funding agencies. Substantial interest in the IDeX program is evident through social networking sites such as Facebook and Twitter, both of which have generated hundreds of unique contacts. Tracking and understanding the development of these networks is essential to determining and sustaining functional networks.

The tools of social network analysis can characterize networks of individuals using reliable and valid data collection instruments and analysis tools (Wasserman and Faust 1994). Social (or organizational) network analysis can be used to gather information on interactions between individuals through selected modes and analyzing the information by using calculated network measures and network maps. Modes are the context for interaction. For example, two faculty members may interact over research, teaching, personal friendship, etc. Multiple network measures exist to investigate both individuals and networks. Most relevant to this project are "connectedness," "indegree," "outdegree," "betweenness," and "prestige." Indegree is the number of people who go to an individual, outdegree is the number of people that an individual goes to for information, and connectedness is the sum of the two. Betweenness is the probability that an individual lies on the shortest path between two other people. For example, considering the mode of real estate, a broker would have a high betweenness value between buyers and sellers. Prestige is an indicator of the ratio of indegree to outdegree. A prestigious member of a network would have a large proportion of people seeking information from him or her, compared with the number of people sought out by that person for information. These network values will be used to identify who the key players are in the IDeX network, the individuals most sought for information, and determine how well individuals within the network are connected.

Network maps can reveal aspects of the network that are not evident in network measure. Network maps consist of boxes, representing people, and arrows, representing connections between people. Network maps can be customized to differentiate among types or characteristics of actors by using shapes and colors. For example, students can be displayed as squares and faculty as circles.

Network data will be collected annually to examine the emergence of a network centered on IDeX and, using network measures and maps, the growth and reasons for growth of the network analyzed to determine project success and how to continue to foster this development. Network data from the first year is somewhat nonconclusive, owing primarily to it being a pilot year and the lack of a substantial focus on integrating numerous practitioners and faculty from other disciplines. In other words, no substantial growth in the formal IDeX network occurred in the first year. Therefore, it is not included in this paper.

Diffusion of Innovations

The majority of historical efforts at changing curricula are short-lived and implemented on the basis of personal experience and anecdotal evidence. Although some succeed, utilization of a valid and reliable theory-based framework for adoption of innovations is a more effective way to ensure success. The difference between a researcher's or instructor's knowledge of a solution and the holistic application of the solution is vast. Diffusion of innovation (DI) (Rogers 2003) provides theory-based perspectives for assessing the efficacy of an innovation into the engineering curriculum. Faculty and practitioner involvement over the course of the project are evaluated using DI theory.

Diffusion of innovation provides a general diffusion model to describe the adoption of an innovation by members of a social system. DI theory is developed on the basis of Rogers' broad definition of an innovation as "an idea, practice, or object that is perceived as new by an individual" (Rogers 2003). It also includes a model to describe an individual's innovation-decision process that consists of five stages that an individual progresses through over time as they decide to adopt or reject and innovation: (1) knowledge, (2) persuasion, (3) decision, (4) implementation, and (5) confirmation. The knowledge stage is the beginning of the innovation-decision process and occurs when the individual is first exposed to the innovation and begins to obtain information about it. The persuasion stage includes the time as the individual forms an opinion, or attitude, about the innovation by actively seeking out information about the innovation. This attitude is developed on the basis of the individual's perceptions of the innovation and is related to the innovation attributes that have been defined by Rogers. The decision stage occurs when the individual begins to make a choice on whether to adopt or reject the innovation. At this point, the adoption-decision process is still about thinking and deciding rather than acting. The implementation stage represents the move into acting on the choice to adopt an innovation and put it to use. The individual will begin to participate in activities that include the innovation, or otherwise begin to incorporate the innovation into their existing practices. The confirmation stage occurs when the individual who has chosen to adopt an innovation seeks reinforcement of this decision among his or her peer groups.

A significant piece of the innovation-decision process occurs during the persuasion stage when the individual begins to form an opinion about the innovation. Rogers identified five perceived attributes of innovations that have large effects on the formation of this opinion and therefore on the success of innovation adoption. This is developed on the basis of the individual's perceptions of the innovations attributes, and not the innovation's attributes as "classified objectively by experts" (Rogers 2003). The five categories proposed by Rogers as important to adoption of an innovation are (1) relative advantage, (2) observability, (3) trialability, (4) compatibility, and (5) complexity. In each case the "actual" quality of each of these five categories related to the innovation process is not important, but rather it is the "relative" quality that is. Relative advantage describes how potential adopters perceive the innovation as better than others, usually as economic profitability, social prestige, or some other form of life improvement. "Observability is closely related to relative advantage. It describes the degree to which an innovation's benefits are visible to potential adopters. Trialability describes how conveniently an innovation can be partially adopted. For example, computer software usually has high trialability because potential adopters can use a free trial version and observe the relative advantages and ease of use before committing to adopting it by purchasing a full version. Compatibility describes how well an innovation's purpose and use integrate with an individual's values and norms. An innovation's complexity can be seen as the level of difficulty it presents to potential adopters to visualize its application and use. Again, this does not relate to the actual

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complexity of the innovation but the perception of how complex it is compared with existing correlates.

Information concerning the adoption of and participation with the IDeX course will be collected through interviews with faculty and affiliated practitioners. The interviewers will ask questions regarding the adopter's behaviors and thoughts during their innovation-decision process such as how they heard about the IDeX course, where they subsequently obtained more information about the course, who they talked to about the course, and how they began to get involved with the course. The questions will also focus on the adopter's perceptions of the IDeX course with respect to the innovation attributes as defined within DI theory. These data will be used to understand how faculty and practitioners have become involved with the IDeX course and their beliefs and perceptions of the course. This understanding can then be used in the development of the course to further encourage participation in the course by faculty and practitioners.

Combined efforts of examining student and faculty learning, developing a functional network, and examining attitudes of IDeX through DI theory will pave the way for the development of a transferable model of authentic and context-rich education of interdisciplinary students on sustainable design principles.

Conclusions

For professionals responsible for the design, construction, and performance of society's infrastructure, it is crucial to face the challenges associated with extending limited natural resources to meet the needs of a growing population with greater expectations for their quality of life. Given the current state of energy and water use, existing models for design do not meet this challenge. Tremendous changes will take place in these professions over the next 25–50 years, which is exactly the career spans for emerging professionals. Academics must evolve the educational systems to prepare young professionals with the tools to evolve the systems. Most important of these many tools are:

- A knowledge of how their individual design decisions influence a wide array of resource uses over the lifetime of a built design;
- An understanding of how designs are influenced by a plethora of professionals working from different viewpoints and toward different goals; and
- A greater understanding and appreciation for the process, language, and value that the different disciplines bring to the design process.

The overall goal is to produce young professionals who are better skilled in communicating and working with colleagues than those of previous generations. Attaining this goal will make them better equipped to respond to future design challenges.

The Integrated Design Experience, a capstone course for undergraduate and graduate students that provides the opportunity to interact with faculty and practitioners on real projects with challenging needs in sustainability, was designed and developed to improve the academic system for training engineers and design and construction professionals. The IDeX is in its infancy but holds real promise for the education of future professionals. The writers envision that the IDeX program can be expanded using two approaches: (1) implementing a similar model at other universities and (2) developing collaborative networks of students and faculty from other universities with a variety of specialized expertise. There is substantial promise to ally groups of students and faculty from multiple universities toward a single project facilitated through online interactions. This approach may be most valuable in satisfying the need for a wide variety of specialized expertise to drive infrastructure performance to higher levels. Finally, there is anticipation for a paradigm shift in the ways engineers and architects interact in the future as a result of IDeX-style training.

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